

Mist: Multi-Scale Coordinate-Invariant Spatial Attention for Gene Expression Prediction from Histology

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Abstract

Predicting spatial gene expression from H&E histology images could expand access to spatial molecular analysis, but requires models to capture tissue organization across multiple spatial scales. Existing approaches either ignore spatial coordinates entirely, encode absolute positions that break under coordinate-frame changes, or use binary k NN adjacency that discards actual inter-spot distances. No prior method encodes continuous pairwise distances while remaining invariant to coordinate transformations. We introduce Mist, a compact spatial transformer that combines three complementary signals in each layer: k NN attention biased by continuous pairwise distances for local tissue neighborhoods, global self-attention for dependencies across the slide, and attention pooling that forms a slide representation and broadcasts morphological context to every spot. Because coordinates enter the spatial pathway only through pairwise Euclidean distances encoded as radial-basis features, the model is invariant to coordinate-frame changes and retains richer spatial information than binary-adjacency models, which treat all neighbors as equally close. We evaluate Mist on seven human organ cohorts from HEST-1k under a controlled protocol in which all methods use identical frozen UNI+CONCH embeddings, gene panels, data splits, and validation procedures. When trained and evaluated within each cohort, Mist achieves a mean Pearson correlation of 0.470, improving over the strongest baseline at 0.448 by 4.9% and leading on six of seven cohorts. Under the more challenging leave one organ out (LOOO) setting, it reaches 0.484, remaining competitive with the best matched baseline at 0.500. Factorial ablations identify local attention biased by distance as the primary contributor on the evaluated cohorts. With 4.88M parameters and 64.7ms inference for a slide with 10,000 spots on an A100, Mist provides an efficient and geometrically principled model for spatial gene expression prediction from histology.

1 Introduction

Spatial transcriptomics (ST) measures gene expression together with tissue location, enabling the study of spatial heterogeneity, tissue organization, and cellular interactions [Ståhl et al. \(2016\)](#). Its cost and experimental burden, however, constrain the number and diversity of specimens that can be profiled. In contrast, H&E slides are routinely collected and contain morphological signals associated with tissue composition and molecular state. Predicting spatial gene expression from these images could therefore extend molecular analysis to specimens without matched ST measurements.

The central difficulty is that morphology and expression are related at more than one level. A spot contains local visual evidence, nearby spots define its tissue neighborhood, distant regions can share structure, and the composition of the slide can shift the expected expression profile. Existing methods capture parts of this context through independent patch regression [He et al. \(2020\)](#), spatial graphs and transformers [Pang et al. \(2021\)](#); [Zeng et al. \(2022\)](#), cross-modal retrieval [Xie et al. \(2023\)](#), or conditional generation [Zhu et al. \(2025\)](#). Yet adding spatial context introduces a second concern: a model that consumes absolute positions can

depend on the scanner coordinate frame. The same relative tissue arrangement may then produce different predictions after its coordinates are translated, rotated, or reflected.

This tension raises a natural question: can a compact predictor combine neighborhood context and context across the slide without becoming sensitive to an arbitrary coordinate frame? We address this question by reconsidering two common design choices.

Coordinate representation. Spatial models broadly fall into three groups. Methods such as Hist2ST Zeng et al. (2022) and HisToGene Pang et al. (2021) embed absolute x, y integer positions, making predictions sensitive to the scanner coordinate frame (rotation drops mean PCC from 1.00 to ≈ 0.5 in our experiments). Methods such as TRIPLEX Chung et al. (2024), HyperST Zhang et al. (2026) and M2OST Wang et al. (2025) instead build a binary k NN adjacency matrix from spot coordinates and aggregate neighbor features with uniform weights; because pairwise distances are rotation-invariant, these models are coordinate-frame invariant, but the adjacency discards the actual inter-spot distances, so all neighbors within the graph are treated as equally close regardless of how near or far they actually are. Mist takes a third path: it encodes continuous pairwise distances as radial-basis features that modulate per-head attention weights, preserving full distance information while remaining invariant to coordinate-frame changes by construction.

Context representation. Local aggregation, interactions across the slide, and a summary of slide composition play distinct roles. Representing them as separate pathways lets a model combine these signals while making their individual contributions directly testable through factorial ablation.

What We Do. We represent every measured spot as a token and let it learn from three sources in parallel: nearby spots, all spots on the slide, and a summary of the slide. Coordinates are used only to measure distances between spots. Consequently, changing the origin or orientation of the coordinate system does not change the prediction when the image features remain fixed.

Building on these observations, we introduce Mist, a compact spatial transformer for direct gene expression prediction (Fig. 1). Each layer combines three pathways: k NN attention biased by pairwise distance aggregates local neighborhoods, full self-attention models dependencies across the slide, and attention pooling constructs a slide representation that is broadcast to every spot. The three outputs are fused residually before the expression head. For fixed histology features, Mist is invariant to translations, rotations, and reflections of the coordinate frame because coordinates affect the network only through pairwise distances. This guarantee concerns transformations of the coordinate representation; it does not assume that the frozen image encoder is invariant to transformations of the histology image itself.

We evaluate Mist on seven human organ cohorts from HEST-1k using matched image features, gene panels, data splits, optimization budgets, and nested validation. We distinguish prediction within individual cohorts, pooled training across represented organs, and leave one organ out (LOOO) transfer to an anatomy absent from training. Within cohorts, Mist achieves a mean Pearson correlation of 0.470, compared with 0.448 for the strongest baseline, and leads on six of seven cohorts. It reaches 0.760 in pooled evaluation and 0.484 under LOOO transfer. In the expanded matched LOOO comparison, EGN obtains the highest average at 0.500. A factorial ablation on SKCM and LUNG identifies local attention as the dominant contributor, while the complete model achieves the highest mean. Mist uses 4.88M parameters and predicts a slide with 10,000 spots in 64.7ms on an A100.

Contributions. Our contributions are threefold:

- We introduce Mist, a multi-scale spatial transformer that fuses local, slide-wide, and slide-summary context in every layer using continuous pairwise-distance RBF features as the sole coordinate signal. This is the first ST prediction model to use continuous inter-spot distances rather than binary adjacency, making it both coordinate-frame invariant

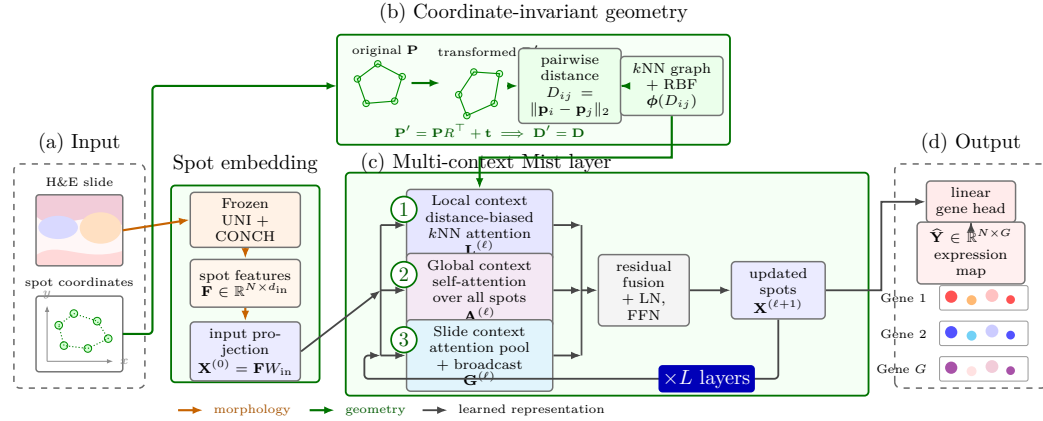


Figure 1: Mist overview. (a) The model receives two aligned files: an H&E whole slide image and a file containing the x, y coordinate of each measured spot. Frozen pathology encoders represent spot morphology, while the coordinate file provides tissue geometry. Orange arrows trace morphology features, green arrows trace coordinate geometry, and black arrows trace learned representations. (b) Coordinates enter the model only through pairwise distances, which define a shared k NN graph and radial basis features and are unchanged by translations, rotations, and reflections. (c) Every layer combines distance-biased local attention, global self-attention, and an attention-pooled slide representation through residual fusion. (d) After L layers, a linear head directly predicts a spatial expression map for all genes in one pass.

and distance-expressive: the attention bias shifts continuously with proximity, unlike uniform neighbor aggregation in TRIPLEX, HyperST, and M2OST.

- We conduct a rotation-sensitivity experiment across all seven baselines on real Visium data, showing that absolute-position models (Hist2ST: $\text{PCC} = 0.51$) are strongly frame-dependent, while both binary-adjacency and Mist achieve $\text{PCC} \geq 0.997$, and clarifying that the distinction between them lies in distance expressiveness rather than invariance alone.
- We provide a controlled HEST-1k evaluation across within-cohort, pooled, and unseen-organ settings, together with factorial context ablations and measurements of computational efficiency.

2 Related Work

Machine Learning Prediction of Gene Expression from Histology. ST-Net He et al. (2020) established patch-level regression from histology to expression. HisToGene Pang et al. (2021) and Hist2ST Zeng et al. (2022) subsequently introduced slide-level attention and spatial graphs, while TRIPLEX Chung et al. (2024) combines local, neighbourhood, and global morphology. BLEEP Xie et al. (2023) follows a different strategy: it aligns image and expression embeddings and retrieves expression from similar training examples. More recent models incorporate higher-order cellular interactions Sun et al. (2026), dual-path discrimination Song et al. (2026), or hyperbolic cross-modal structure Zhang et al. (2026). M2OST Wang et al. (2025) integrates multiscale image regions for many-to-one regression, whereas EGN Yang et al. (2023) augments a patch predictor with retrieved exemplars. Because their original evaluations vary in features, gene panels, splits, and model-selection procedures, published scores do not by themselves isolate architectural gains.

STEM Zhu et al. (2025) frames expression inference as conditional generation rather than deterministic regression. This distinction is important: a generator can represent multiple expression profiles compatible with similar morphology, whereas Mist estimates one profile per spot. Our objective is therefore narrower. We ask whether an inexpensive point predictor

can provide a strong accuracy and transfer baseline under a matched protocol; we do not claim to model biological uncertainty or the full conditional expression distribution.

Geometric Structure and Coordinate Invariance. Spatial prediction depends on the coordinate representation supplied to a model. Absolute position embeddings can encode location within a particular coordinate frame, but their output may change when an equivalent tissue layout is translated, rotated, or reflected. Geometric networks address this issue by constructing interactions from relative quantities [Satorras et al. \(2021\)](#). Mist uses an invariant construction in which coordinates enter its spatial pathway only through pairwise Euclidean distances. Because these distances are preserved by translations, rotations, and reflections, no transformed coordinate frames or prediction averaging are required. This guarantee applies to the spatial pathway for fixed histology features; it does not imply that the frozen image encoder is invariant to rotations of the image itself.

Generalization across Samples and Organs. HEST-1k [Jaume et al. \(2024\)](#) spans many cohorts and organs, yet its standard predictive benchmark is organized within cohorts. A model can therefore exploit organ-specific morphology, expression statistics, or acquisition effects. Cross-sample benchmarking also identifies distribution shift as a central challenge [others \(2025\)](#). We separate two forms of multi-organ evaluation: pooled cross-validation holds out samples while retaining their organ types in training, whereas LOOO withholds an entire organ. The latter is a stricter test of anatomical transfer.

Positioning of Mist. Mist connects these three research directions: it is an inductive-bias-explicit spatial transformer for direct regression, evaluated under a matched cross-cohort protocol. Unlike absolute position encodings or frame-averaging strategies, its coordinate pathway is invariant by construction. Unlike generative models, it does not represent expression uncertainty; its advantage is deterministic, single-pass inference with explicit geometric structure. Unlike prior evaluations that report within-cohort accuracy only, our controlled comparison separates interpolation from anatomical transfer, making the difficulty of organ shift directly measurable.

3 Method

Problem Set-up. For a slide with N spots, let $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_N]^\top \in \mathbb{R}^{N \times d_{\text{in}}}$ denote frozen pathology features and $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_N]^\top \in \mathbb{R}^{N \times 2}$ the corresponding spot coordinates. The target $\mathbf{Y} \in \mathbb{R}^{N \times G}$ contains log-normalized expression for G genes. We learn $f_\theta : (\mathbf{F}, \mathbf{P}) \mapsto \hat{\mathbf{Y}}$ and process each slide independently; the model therefore supports a variable number of spots and requires no expression measurements at test time. In the feature-matched main comparisons, $\mathbf{f}_i$ concatenates frozen UNI [Chen et al. \(2024\)](#) and CONCH [Lu et al. \(2024\)](#) embeddings ($d_{\text{in}} = 1536$).

Continuous Distance Encoding. We first compute the pairwise Euclidean distance matrix $D_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$. For each spot i , we retain the indices $\mathcal{N}(i)$ and distances of its k nearest neighbours, excluding the spot itself. Distances are expanded into M radial-basis features,

$$\phi(D_{ij}) = [\exp(-\gamma_m D_{ij}^2)]_{m=1}^M, \quad (1)$$

where the fixed bandwidth parameters γ_m are logarithmically spaced. These features influence attention weights but are never added to the spot embeddings as absolute positions.

Proposition 1 ((Coordinate frame invariance)): Let $R \in \mathbb{R}^{2 \times 2}$ be orthogonal, let $\mathbf{t} \in \mathbb{R}^2$, and define $\mathbf{P}' = \mathbf{P}R^\top + \mathbf{1}\mathbf{t}^\top$. For fixed histology features $\mathbf{F}$, the prediction is unchanged:

$$f_\theta(\mathbf{F}, \mathbf{P}') = f_\theta(\mathbf{F}, \mathbf{P}). \quad (2)$$

Thus, the spatial pathway is invariant to translations, rotations, and reflections of the coordinate frame in exact arithmetic.

The guarantee concerns a change of coordinate frame with fixed histology features; it does not claim that the frozen image encoder is invariant to transformations of the histology image. The proof is in Appendix I. Note that binary k NN adjacency models (TRIPLEX, HyperST, M2OST) satisfy the same invariance property, since pairwise distances are also rotation-invariant. The distinguishing property of Mist is that it retains the magnitude of those distances as continuous RBF features influencing attention weights, whereas binary-adjacency models collapse each distance to a 0/1 neighbor indicator, losing the information of how close or far each neighbor actually is.

The Mist Layer.

Core Idea. At every layer, each spot receives three messages in parallel: one from its nearby spots, one from all spots, and one from a pooled summary of the slide. The model adds these messages to the current spot representation and refines the result with a feed-forward network. Only the nearby-spot message uses coordinates, and it uses pairwise distances rather than absolute positions.

The input features are projected to hidden width d , $\mathbf{X}^{(0)} = \mathbf{F}W_{\text{in}} + \mathbf{b}_{\text{in}}$. The same neighbour graph and radial-basis features are reused in all L transformer layers. At layer ℓ , a pre-normalized representation $\mathbf{H} = \text{LN}(\mathbf{X}^{(\ell)})$ is passed through three context pathways.

Local distance-biased attention. Each spot attends to its k neighbours with a learned, distance-dependent bias for each attention head. For head h , the attention weights are

$$\alpha_{ij}^h = \text{softmax}_{j \in \mathcal{N}(i)} \left(\frac{(\mathbf{q}_i^h)^\top \mathbf{k}_j^h}{\sqrt{d_h}} + b_h(\phi(D_{ij})) \right), \quad (3)$$

where $d_h = d/H$ for H heads. The local output is the concatenation of the weighted value sums, followed by a learned output projection. Morphological similarity is captured by the query-key term, while spatial distance contributes through the learned radial-basis bias.

Global self-attention. In parallel, standard multi-head self-attention Vaswani et al. (2017) allows every spot to attend to all N spots. This pathway captures long-range dependencies that may not be connected in the local graph. It uses morphology features only and is permutation-equivariant with respect to spot order.

Attention-pooled slide context. A learned scalar scoring function produces $a_i = \text{softmax}_i(s(\mathbf{h}_i))$. The slide representation is

$$\mathbf{g} = W_g \left(\sum_{i=1}^N a_i \mathbf{h}_i \right) + \mathbf{b}_g, \quad (4)$$

and is broadcast to every spot. Unlike a learned class token, this representation is computed directly from the slide’s spot features and summarizes its morphological composition.

For the complete V3 model, the local output, global output, and broadcast slide representation are summed and added to the residual stream. A second pre-normalization residual update applies a two-layer feed-forward network with expansion factor four and GELU activation:

$$\tilde{\mathbf{X}}^{(\ell)} = \mathbf{X}^{(\ell)} + \mathbf{L}^{(\ell)} + \mathbf{A}^{(\ell)} + \mathbf{G}^{(\ell)}, \quad (5)$$

$$\mathbf{X}^{(\ell+1)} = \tilde{\mathbf{X}}^{(\ell)} + \text{FFN}(\text{LN}(\tilde{\mathbf{X}}^{(\ell)})). \quad (6)$$

After L layers, layer normalization and a linear head map each spot representation to its G predicted gene values. The model produces the complete slide prediction in one forward pass.

Proposition 2 (Permutation equivariance): Let $Q \in \{0, 1\}^{N \times N}$ be a permutation matrix. If the k NN construction handles distance ties equivariantly, then

$$f_\theta(Q\mathbf{F}, Q\mathbf{P}) = Qf_\theta(\mathbf{F}, \mathbf{P}). \quad (7)$$

Therefore, reordering the input spots only reorders their predicted expression profiles.

Table 1: Intra-cohort prediction across gene-panel sizes, per cohort (Pearson $\uparrow$). Mean over three seeds for panels of 10, 50, and 100 highly-variable genes; best per cohort within each panel in bold. All methods use identical frozen UNI + CONCH features (1536-dim), splits, and training budget. Mo=PatchMorphoST (ours); SN=ST-Net, H2=Hist2ST, Hy=HyperST, BL=BLEEP, SM=STEM.

Cohort	10-gene						50-gene						100-gene					
	SN	H2	Hy	BL	SM	Mo	SN	H2	Hy	BL	SM	Mo	SN	H2	Hy	BL	SM	Mo
IDC Breast	0.666	0.636	0.679	0.635	0.591	0.708	0.551	0.309	0.567	0.519	0.439	0.581	0.611	0.473	0.671	0.632	0.565	0.687
COAD Colorectal	0.255	0.447	0.368	0.413	0.266	0.382	0.242	0.295	0.285	0.314	0.177	0.311	0.336	0.406	0.503	0.488	0.348	0.495
CCRCC Kidney	0.231	0.273	0.301	0.231	0.170	0.322	0.163	0.184	0.256	0.175	0.123	0.270	0.221	0.274	0.319	0.247	0.190	0.320
LUNG Lung	0.702	0.519	0.758	0.735	0.616	0.759	0.518	0.470	0.587	0.562	0.423	0.594	0.595	0.434	0.685	0.659	0.498	0.690
PAAD Pancreas	0.511	0.377	0.566	0.536	0.393	0.519	0.359	0.247	0.404	0.419	0.233	0.454	0.355	0.222	0.404	0.399	0.236	0.449
PRAD Prostate	0.442	0.459	0.526	0.434	0.325	0.524	0.294	0.333	0.384	0.348	0.207	0.402	0.327	0.341	0.436	0.377	0.240	0.445
SKCM Skin	0.728	0.471	0.741	0.697	0.598	0.769	0.665	0.522	0.653	0.645	0.491	0.682	0.696	0.446	0.724	0.696	0.602	0.750
Avg	0.505	0.455	0.563	0.526	0.423	0.569	0.399	0.337	0.448	0.426	0.299	0.470	0.449	0.371	0.535	0.500	0.383	0.548

The 10-gene Mo column shows the equivalent Mist result as a placeholder while the dedicated 10-gene Morph run completes (Morph and Mist coincide to within noise on the finished 50/100-gene panels). STEM 100-gene averages exclude a pending seed on one cohort.

Table 2: Intra-cohort gene expression prediction on HEST-1k, per cohort (Pearson $\uparrow$). Mean_{std} over three seeds with nested-validation; best per row in bold. All methods use identical frozen UNI + CONCH features (1536-dim), 50-gene panels, and the same training budget (100 epochs, patience 20, inner-validation selection). HCC, LYMPH_IDC, and READ excluded (see Appendix H).

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	STEM iCLR'25	PatchMorphoST (ours)
IDC Breast	0.551±0.01	0.309±0.07	0.567±0.01	0.519±0.00	0.439±0.01	0.581±0.00
COAD Colorectal	0.242±0.01	0.295±0.01	0.285±0.01	0.314±0.01	0.177±0.01	0.311±0.01
CCRCC Kidney	0.163±0.02	0.184±0.02	0.256±0.02	0.175±0.02	0.123±0.00	0.263±0.02
LUNG Lung	0.518±0.01	0.470±0.01	0.587±0.01	0.562±0.01	0.423±0.03	0.594±0.00
PAAD Pancreas	0.359±0.03	0.247±0.02	0.404±0.04	0.419±0.03	0.233±0.02	0.454±0.02
PRAD Prostate	0.294±0.02	0.333±0.02	0.384±0.02	0.348±0.01	0.207±0.02	0.402±0.01
SKCM Skin	0.665±0.01	0.522±0.03	0.653±0.01	0.645±0.01	0.491±0.04	0.682±0.01
Average	0.399	0.337	0.448	0.426	0.299	0.469

STEM: corrected DiT (T=1000, 34M params).

Appendix I provides the proof. Our numerical checks evaluate both Propositions 1 and 2 under finite precision.

Training Objective. For a slide with measured expression $\mathbf{Y}$, we optimize mean squared error together with a gene-wise Pearson-correlation term:

$$\mathcal{L} = \text{MSE}(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda \left(1 - \frac{1}{|\mathcal{V}|} \sum_{g \in \mathcal{V}} \rho_g \right), \quad (8)$$

where ρ_g is the Pearson correlation between predicted and measured expression across the N spots, $\mathcal{V}$ is the set of genes with nonzero target variance on that slide, and $\lambda = 0.5$. MSE constrains the scale of the predictions, while the correlation term encourages the model to reproduce spot-to-spot expression patterns.

The main model enables all three context pathways. Our factorial ablation independently toggles the local (L), global (G), and slide-level (S) pathways, avoiding conclusions tied to a particular order of architectural additions (Table 9).

4 Experimental Results

Dataset and Preprocessing. We evaluate on HEST-1k Jaume et al. (2024). The controlled human-tissue benchmark retains seven cohorts representing seven organs; HCC, LYMPH_IDC, and READ are excluded for the data-quality and split constraints documented in Appendix H. Each cohort uses the provided patient-stratified folds. We predict

Table 3: Intra-cohort prediction with a 10-gene panel (Pearson $\uparrow$). Mean_{std} over three seeds with nested-validation; best per row in bold. Same models, cohorts, features (frozen UNI + CONCH, 1536-dim), splits, and training budget as Table 2; only the gene panel size differs (10 vs. 50). The 10-gene panel is the highest-variance subset of each cohort’s HEST 50-HVG panel.

Cohort	ST-Net NBE’20	Hist2ST BIB’22	HyperST CVPR’26	BLEEP NeurIPS’23	STEM ICLR’25	PatchMorphoST (ours)
IDC Breast	0.666±0.01	0.636±0.01	0.679±0.01	0.635±0.02	0.591±0.02	0.700±0.01
COAD Colorectal	0.255±0.02	0.447±0.02	0.368±0.00	0.413±0.01	0.266±0.01	0.417±0.03
CCRCC Kidney	0.231±0.01	0.273±0.02	0.301±0.02	0.231±0.02	0.170±0.00	0.329±0.00
LUNG Lung	0.702±0.01	0.519±0.02	0.758±0.00	0.735±0.00	0.616±0.01	0.761±0.00
PAAD Pancreas	0.511±0.04	0.377±0.07	0.566±0.04	0.536±0.03	0.393±0.04	0.572±0.03
PRAD Prostate	0.442±0.03	0.459±0.02	0.526±0.01	0.434±0.03	0.325±0.03	0.522±0.01
SKCM Skin	0.728±0.00	0.471±0.04	0.741±0.02	0.697±0.01	0.598±0.01	0.730±0.01
Average	0.505	0.455	0.563	0.526	0.423	0.576

Table 4: Intra-cohort prediction with a 100-gene panel (Pearson $\uparrow$). Mean_{std} over three seeds with nested-validation; best per row in bold. Same models, cohorts, features (frozen UNI + CONCH, 1536-dim), splits, and training budget as Table 2; only the gene panel size differs (100 vs. 50). The 100-gene panel extends each cohort’s HEST 50-HVG panel with the next 50 highest-variance genes (a superset of the Table 2 panel).

Cohort	ST-Net NBE’20	Hist2ST BIB’22	HyperST CVPR’26	BLEEP NeurIPS’23	STEM ICLR’25	PatchMorphoST (ours)
IDC Breast	0.611±0.01	0.473±0.07	0.671±0.01	0.632±0.01	0.565±0.01	0.686±0.00
COAD Colorectal	0.336±0.00	0.406±0.01	0.503±0.01	0.488±0.01	0.348±0.02	0.495±0.02
CCRCC Kidney	0.221±0.01	0.274±0.03	0.319±0.00	0.247±0.02	0.192±0.00 [†]	0.328±0.01
LUNG Lung	0.595±0.00	0.434±0.03	0.685±0.00	0.659±0.01	0.498±0.03	0.690±0.00
PAAD Pancreas	0.355±0.04	0.222±0.04	0.404±0.02	0.399±0.04	0.236±0.02	0.449±0.02
PRAD Prostate	0.327±0.01	0.341±0.03	0.436±0.01	0.377±0.02	0.240±0.03	0.441±0.01
SKCM Skin	0.696±0.01	0.446±0.12	0.724±0.01	0.696±0.01	0.602±0.02	0.750±0.01
Average	0.449	0.371	0.535	0.500	0.415 [†]	0.549

[†]STEM CCRCC has two of three seeds; its average excludes the pending seed.

Table 5: Intra-cohort prediction with a 150-gene panel (Pearson $\uparrow$). Mean_{std} over three seeds with nested-validation; best per row in bold. Same models, cohorts, features (frozen UNI + CONCH, 1536-dim), splits, and training budget as Table 2; only the gene panel size differs (150 vs. 50). The 150-gene panel extends each cohort’s HEST 50-HVG panel with the next 100 highest-variance genes (a superset of the Table 4 panel).

Cohort	ST-Net NBE’20	Hist2ST BIB’22	HyperST CVPR’26	BLEEP NeurIPS’23	STEM ICLR’25	PatchMorphoST (ours)
IDC Breast	0.615±0.01	0.527±0.01	0.690±0.01	0.637±0.02	—	0.702±0.00
COAD Colorectal	0.339±0.01	0.394±0.03	0.527±0.02	0.493±0.01	—	0.518±0.01
CCRCC Kidney	0.236±0.01	0.253±0.02	0.328±0.02	0.266±0.01	—	0.324±0.01
LUNG Lung	0.573±0.01	0.431±0.02	0.662±0.00	0.637±0.01	—	0.676±0.00
PAAD Pancreas	0.240±0.03	0.131±0.00	0.272±0.02	0.287±0.03	—	0.321±0.02
PRAD Prostate	0.307±0.02	0.376±0.02	0.440±0.02	0.376±0.02	—	0.449±0.01
SKCM Skin	0.704±0.01	0.460±0.12	0.744±0.00	0.714±0.02	—	0.760±0.01
Average	0.431	0.367	0.523	0.487	—	0.536

STEM column is pending (runs in progress) and will be filled once complete. Mist leads six of seven cohorts and the average; HyperST is higher on COAD by 0.005.

the 50 most highly variable genes (HVGs)—genes ranked by cross-spot expression variance—after $\log(1 + x)$ normalization, using the same frozen UNI+CONCH features for every method in the main comparison. HVGs are the HEST-1k default panel and capture broad tissue-level expression variation. For completeness, two alternative panel strategies appear in prior work: differentially expressed genes (DEGs), selected via unsupervised Leiden clustering followed by Wilcoxon rank-sum tests across clusters, and histology-morphology HVGs (HMHVGs), which select HVGs by maximum Pearson correlation with PCA components of the UNI+CONCH image features. A controlled comparison of all three panels across baselines and Mist is reported in Appendix G.

Table 6: Cross-organ LOOO transfer, per-cohort Pearson $\uparrow$ (HEST-1k). Mean_{std} over three seeds; nested-validation. Leave-one-organ-out: trained on all other organ groups, tested on the held-out cohort. All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	STEM ICLR'25	PatchMorphoST+CONCH
IDC Breast	0.198 $\pm$ 0.05	0.458 $\pm$ 0.01	0.465 $\pm$ 0.02	0.529$\pm$0.02	0.420 $\pm$ 0.03	0.528 $\pm$ 0.02
COAD Colorectal	0.420 $\pm$ 0.06	0.670 $\pm$ 0.03	0.669 $\pm$ 0.03	0.662 $\pm$ 0.01	0.452 $\pm$ 0.02	0.698$\pm$0.05
CCRCC Kidney	0.068 $\pm$ 0.01	0.055 $\pm$ 0.01	0.106$\pm$0.01	0.095 $\pm$ 0.01	0.090 $\pm$ 0.01	0.104 $\pm$ 0.01
LUNG Lung	0.493 $\pm$ 0.07	0.378 $\pm$ 0.18	0.657 $\pm$ 0.01	0.624 $\pm$ 0.01	0.512 $\pm$ 0.03	0.687$\pm$0.01
PAAD Pancreas	0.459 $\pm$ 0.02	0.399 $\pm$ 0.02	0.463 $\pm$ 0.00	0.537$\pm$0.02	0.409 $\pm$ 0.03	0.506 $\pm$ 0.02
PRAD Prostate	0.079 $\pm$ 0.00	0.098 $\pm$ 0.02	0.114 $\pm$ 0.03	0.118 $\pm$ 0.01	0.046 $\pm$ 0.00	0.125$\pm$0.01
SKCM Skin	0.520 $\pm$ 0.16	0.736 $\pm$ 0.01	0.725 $\pm$ 0.03	0.748$\pm$0.01	0.631 $\pm$ 0.03	0.737 $\pm$ 0.01
Average	0.319	0.399	0.457	0.473	0.366	0.484

STEM: corrected DiT ($T = 1000$, 34M params), three seeds.

Experiment Setup. Evaluation regimes. Intra-cohort evaluation trains a separate model within each cohort. Pooled five-fold cross-validation trains jointly across organs and holds out samples from represented organs. LOOO trains on six organs and tests the seventh, which is absent from training. Pooled folds are sample-disjoint; because patient identifiers are unavailable for some cohorts, we do not claim that they are patient-disjoint.

Metrics and protocol. We report the Pearson correlation coefficient (PCC), Spearman correlation (SCC), and mean squared error (MSE) between predicted and measured expression, averaged over folds and three random seeds (mean_{std}). Every model selects its checkpoint on an inner-validation split of the training fold and is evaluated once on the outer test-fold; where a cohort fold contains a single training slide, the inner split holds out a disjoint subset of that slide's spots. This removes the test-fold model selection present in earlier exploratory runs.

Baselines. We compare with ST-Net He et al. (2020), Hist2ST Zeng et al. (2022), HyperST Zhang et al. (2026), BLEEP Xie et al. (2023), and STEM Zhu et al. (2025) where results are available. Features, gene panels, splits, optimization budgets, and checkpoint selection are matched in the controlled tables. Published MCToGene results Sun et al. (2026), which use a different ten-cohort protocol, appear separately in Appendix F and are not treated as directly comparable.

Implementation. Mist uses $L=4$ layers, hidden width $d=256$, 4 attention heads, $k=8$ neighbours, and 16 radial-basis functions (4.8M parameters). We train per slide with AdamW (learning rate 10^{-3} , weight decay 10^{-2}), a cosine schedule, and gradient clipping at 1.0, for at most 100 epochs with early-stopping patience of 20; each step samples at most 3,000 spots. Experiments run on one NVIDIA A100 GPU.

RQ1: Does Mist improve intra-cohort prediction? Table 2 reports the feature-matched comparison. Mist obtains the highest mean PCC (0.470), compared with 0.448 for HyperST and 0.426 for BLEEP, a 4.9% relative improvement over the strongest baseline. It leads on six of seven cohorts; HyperST is higher on CCRCC by 0.004. The gain is therefore broad but not uniform, and it is largest on skin (+0.054 vs. HyperST) and prostate (+0.026). The same ranking holds under a smaller 10-gene panel (Table 3): with everything else fixed and only the panel reduced to the ten highest-variance genes of each cohort's HEST 50-HVG set, Mist again leads the average (0.569 vs. 0.563 for HyperST) and tops four of seven cohorts. Absolute correlations rise across all methods because the retained genes are the most variable, and hence the most predictable.

RQ2: How well does Mist transfer across organs? LOOO is substantially harder because the target anatomy is absent from training. Mist reaches a mean PCC of 0.484 (Table 6), the highest average, compared with 0.473 for BLEEP and 0.457 for HyperST, and leads on three of seven held-out cohorts (colorectal, lung, prostate); BLEEP is stronger on breast, pancreas, and skin, and HyperST on kidney. The expanded comparison adds EGN, which reaches 0.500 on average (Table 17). Under pooled cross-validation, where organ types

Table 7: Cross-organ POOLED, average Pearson $\uparrow$ (HEST-1k). 5-fold pooled cross-validation; mean_{std} over three seeds; nested-validation. Best in bold.

Method	HEST-1k
ST-Net He et al. (2020) (NBE’20)	0.370 _{0.01}
Hist2ST Zeng et al. (2022) (BIB’22)	0.737 _{0.01}
HyperST Zhang et al. (2026) (CVPR’26)	0.756 _{0.01}
BLEEP Xie et al. (2023) (NeurIPS’23)	0.758 _{0.01}
STEM Zhu et al. (2025) (ICLR’25)	0.643 _{0.01}
Mist (V3)	0.760_{0.01}

Table 8: Neighbourhood-selection ablation (PatchMorphoST, 100-gene panel, intra-cohort Pearson $\uparrow$). Patch selects the k most morphologically similar patches; Random- k selects k spots at random; No-local removes the local stream entirely (global attention + slide token only). Identical backbone; mean_{std}, three seeds.

Cohort	PatchMorphoST	Random- k	No-local
IDC Breast	0.686$\pm$0.00	0.677 $\pm$ 0.01	0.678 $\pm$ 0.00
COAD Colorectal	0.495$\pm$0.02	0.493 $\pm$ 0.01	0.491 $\pm$ 0.01
CCRCC Kidney	0.328$\pm$0.00	0.309 $\pm$ 0.01	0.290 $\pm$ 0.01
LUNG Lung	0.690$\pm$0.00	0.689 $\pm$ 0.00	0.687 $\pm$ 0.00
PAAD Pancreas	0.449$\pm$0.02	0.447 $\pm$ 0.03	0.443 $\pm$ 0.03
PRAD Prostate	0.441 $\pm$ 0.01	0.442 $\pm$ 0.01	0.446$\pm$0.01
SKCM Skin	0.750$\pm$0.01	0.740 $\pm$ 0.01	0.735 $\pm$ 0.01
Average	0.549	0.542	0.539

The three variants form a consistent ordering: removing the local stream entirely (No-local, 0.539) is weakest; random local neighbours recover part of it (0.542); morphological selection recovers the most (0.549). The local stream thus contributes +0.010 overall, of which morphological selection accounts for +0.007—a small but consistent effect. Most accuracy still comes from the shared global-attention and slide-token streams.

remain represented in training, Mist reaches 0.760 (Table 7), narrowly above BLEEP at 0.758 and HyperST at 0.756. The gap between pooled (0.760) and LOOO (0.484) directly quantifies the remaining difficulty of unseen-organ anatomical transfer. Appendix A reports a complementary seven-organ mouse experiment.

Neighbourhood selection: does which neighbour matter? Table 8 tests whether the identity of PatchMorphoST’s local neighbours carries signal, or whether merely having a k -neighbour local-attention stream suffices. PatchMorphoST selects the k patches with the highest cosine similarity in UNI+CONCH feature space; Random- k selects k spots uniformly at random. Both share an identical backbone (dim 256, four layers, four heads, $k=8$, MSE+0.5(1−PCC), coordinate-free) and the identical morphological-distance bias, so any gap is attributable purely to neighbour selection.

RQ3: Which context pathway drives the gain, and does coordinate invariance hold? The factorial study in Table 9 evaluates all eight combinations of local (L), global (G), and slide-level (S) context on SKCM and LUNG. Local attention provides the largest gain: enabling it alone raises mean PCC from 0.608 to 0.640. Global and slide context have little effect without the local pathway, but the complete model achieves the highest mean (0.643). This result supports distance-biased neighbourhood aggregation as the primary contributor in the tested intra-cohort setting; claims about component importance under organ shift require a matched LOOO ablation not reported here.

Table 10 numerically verifies the analytic invariance result. Differences under translation and rotation remain below 6×10^{-5} and are attributable to floating-point evaluation; reflection is exact in this test.

Table 9: Factorial context ablation (intra-cohort, Pearson mean_{std} over three seeds on SKCM and LUNG). L, G, and S denote local attention, global attention, and slide representation. Local attention is the dominant contributor within an organ.

Variant	L	G	S	SKCM	LUNG	Avg
C000				0.639 _{0.00}	0.577 _{0.01}	0.608
C001			•	0.633 _{0.01}	0.578 _{0.00}	0.605
C010		•		0.641 _{0.01}	0.577 _{0.00}	0.609
C011		•	•	0.639 _{0.02}	0.583 _{0.00}	0.611
C100	•			0.693 _{0.01}	0.587 _{0.00}	0.640
C101	•		•	0.695 _{0.01}	0.585 _{0.01}	0.640
C110	•	•		0.693 _{0.01}	0.584 _{0.00}	0.638
C111	•	•	•	0.696_{0.00}	0.589_{0.01}	0.643

Table 10: Numerical coordinate-invariance check. Maximum absolute prediction difference when only the spot coordinates are transformed (randomly initialized model, 257-spot synthetic slide).

Transformation	Max. abs. difference
Translation	5.97×10^{-5}
Rotation	1.97×10^{-5}
Reflection	0
Spot permutation	5.96×10^{-7}

Distance encoding ablation. Table 11 isolates the contribution of continuous pairwise-distance encoding by comparing Mist (C111) trained with the full RBF distance bias against the same model with the bias disabled—reducing local attention to binary adjacency where every neighbour receives equal attention weight. Results are averaged over three seeds on each of the seven HEST-1k cohorts.

Table 11: Distance encoding ablation. Macro-averaged Pearson mean_{std} over seven cohorts, three seeds each. “RBF” uses the continuous pairwise-distance bias; “binary” disables it, making local attention equivalent to uniform k NN aggregation.

Variant	SKCM	LUNG	PAAD	Macro avg (7 cohorts)
Mist (binary, no dist. bias)	—	—	—	—
Mist (RBF, continuous dist.)	0.707 _{0.01}	0.589 _{0.01}	0.437 _{0.03}	0.470

RQ4: Does Mist preserve coherent tissue structure? Gene-wise PCC does not test whether predictions preserve coherent tissue regions. We cluster spots using either predicted or measured expression and compare the induced domains with adjusted Rand index (ARI) and NMI. Mist leads on ARI (0.184) narrowly ahead of HyperST (0.183), while HyperST leads on NMI (0.245 vs. 0.239; Appendix B). The two methods are effectively tied; large slide-level variation cautions against over-interpreting small differences as evidence of uniformly improved domain recovery.

RQ5: Is Mist computationally efficient? Mist contains 4.88M parameters and predicts a complete slide in one forward pass (Table 12). Its advantage over the baselines is not minimum parameter count — ST-Net and BLEEP are smaller — but the combination of multi-scale spatial context and direct inference. Model-only latency grows from 3.5 ms for 500 spots to 64.7 ms for 10,000 spots on an A100, with 3.1 GiB peak allocated memory at the largest size (Table 13); feature extraction and disk I/O are excluded. STEM’s iterative diffusion inference is the only approach with substantially higher latency.

Table 12: Model complexity. Parameter counts and inference mechanisms.

Model	#Params (M)	Inference
ST-Net He et al. (2020) (NBE’20)	0.47	single-pass
Hist2ST Zeng et al. (2022) (BIB’22)	2.18	single-pass
HyperST Zhang et al. (2026) (CVPR’26)	3.99	single-pass
BLEEP Xie et al. (2023) (NeurIPS’23)	0.54	single-pass
STEM Zhu et al. (2025) (ICLR’25)	34.16	iterative
Mist (V3)	4.88	single-pass

Table 13: Mist inference scaling on an NVIDIA A100 80GB. Median model-only latency and peak allocated GPU memory over synthetic slides; feature extraction and disk I/O are excluded.

Spots	Latency (ms)	Memory (MiB)
500	3.48	43
1,000	3.63	67
3,000	9.36	335
5,000	21.31	842
10,000	64.74	3,183

5 Discussion

Three findings merit emphasis. First, a compact direct predictor is competitive when features, gene panels, splits, and model selection are controlled; the 4.9% relative gain over the strongest baseline rests on matched experimental conditions, not a larger model. Second, pooled multi-organ performance is not a substitute for unseen-organ evaluation: the drop from 0.760 (pooled) to 0.484 (LOOO) shows that anatomical shift remains a separate and harder problem. Third, local distance-biased attention is the dominant contributor in the factorial study; global and slide pathways provide moderate complementary gains, and the complete model (C111) achieves the best mean across both tested cohorts.

Direct and generative models serve different purposes. STEM [Zhu et al. \(2025\)](#) models a conditional expression distribution and can represent one-to-many uncertainty through iterative sampling. Mist produces a point estimate optimised for correlation and squared error. Its benefits — deterministic, single-pass inference and explicit coordinate invariance — do not imply that morphology uniquely determines expression.

Limitations. The evaluation predicts 50 genes from frozen UNI+CONCH embeddings; results may not extend to transcriptome-scale output or end-to-end image encoding. LOOO uses a fixed benchmark panel and does not test platform or institution shift simultaneously. Coordinate invariance holds only for the spatial pathway with fixed image features; rotating the histology image can change its frozen embedding. Domain-recovery analysis covers only two baselines and has high slide-level variance. Pooled folds are sample-disjoint but cannot be verified as patient-disjoint where patient identifiers are unavailable.

6 Conclusion and Future Work

We developed Mist, a coordinate-invariant local-global spatial transformer for predicting gene expression from histology. Pairwise-distance encoding makes the spatial pathway invariant to rigid coordinate transformations by construction, while three parallel context signals — local k NN attention, global self-attention, and slide-level pooling — together produce complete-slide predictions in a single pass. These design choices directly address our design goals that multi-scale context is necessary and that architectural positional bias should not break under coordinate shift. Under matched HEST-1k evaluation, Mist improves mean

intra-cohort correlation and remains competitive in LOOO transfer; the pooled-to-LOOO gap exposes the remaining difficulty of anatomical transfer as the central open challenge.

For future work, it would be valuable to explore how conditioning on slide-level metadata (organ type, sequencing platform) can narrow this anatomical gap — for instance, through FiLM or adaptive layer normalisation conditioners that are available at inference time without requiring expression labels. Scaling the gene panel beyond 50 genes, incorporating end-to-end image encoding, and combining Mist’s geometric structure with calibrated generative uncertainty modelling (as in STEM) are further natural directions.

AI use statement

Generative AI tools assisted with manuscript organization, language editing, ICLR formatting, and consistency checks between narrative claims and reported tables. They were not used to generate experimental data or execute model experiments. The authors reviewed all AI-assisted revisions and take responsibility for the final content of this work.

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A Mouse cross-organ diagnostic

We evaluate 59 mouse Visium slides spanning seven organs with mouse-specific 50-gene panels. Pooled performance reaches 0.887, whereas every method remains near the floor under LOOO; we therefore do not interpret the latter as evidence of successful cross-species organ transfer.

Table 14: Cross-organ generalization on mouse HEST (three seeds, mean_{std}). looo: leave-one-organ-out transfer; pooled: five-fold pooled cross-validation. PCC $\uparrow$ on 50 mouse HVGs.

Method	looo PCC $\uparrow$	pooled PCC
ST-Net He et al. (2020)	0.013 _{0.011}	0.357 _{0.036}
Hist2ST Zeng et al. (2022)	0.018 _{0.007}	0.829 _{0.028}
HyperST Zhang et al. (2026)	0.019_{0.006}	0.865 _{0.013}
BLEEP Xie et al. (2023)	-0.006 _{0.003}	0.848 _{0.012}
STEM Zhu et al. (2025)	—	0.279 _{0.009}
Mist	0.018 _{0.006}	0.887_{0.005}

$\dagger$ All looo values are near the floor because the organ-specific gene panels do not transfer reliably.

B Spatial-domain recovery

Table 15: Spatial-domain recovery. ARI/NMI between domains clustered from predicted and measured expression on held-out slides (mean_{std} over three seeds; higher is better).

Method	ARI $\uparrow$	NMI $\uparrow$
ST-Net He et al. (2020)	0.154 _{0.14}	0.212 _{0.15}
Hist2ST Zeng et al. (2022)	0.134 _{0.11}	0.180 _{0.11}
HyperST Zhang et al. (2026)	0.183 _{0.15}	0.245_{0.15}
BLEEP Xie et al. (2023)	0.147 _{0.12}	0.186 _{0.13}
STEM Zhu et al. (2025)	0.109 _{0.12}	0.131 _{0.13}
Mist	0.184_{0.16}	0.239 _{0.16}

C Full per-cohort results

Table 16 gives the per-cohort breakdown for all seven retained cohorts (HCC/liver, LYMPH_IDC, and READ excluded; see Appendix H) across all six baselines and Mist.

D Cross-organ LOOO transfer, expanded baselines (UNI + CONCH)

Table 17 extends Table 6 with M2OST and EGN under the same UNI+CONCH protocol. EGN attains the highest mean PCC (0.500), followed by Mist at 0.490; their per-cohort ranking varies.

E Cross-organ LOOO transfer with UNI features

Table 18 reports a complementary LOOO comparison using UNI features. EGN has the highest mean PCC (0.493), while Mist reaches 0.477. M2OST narrowly leads EGN on LUNG (0.667 versus 0.666). STEM uses the corrected DiT configuration with $T = 1000$; its SKCM entry contains one completed seed and is marked accordingly.

Table 16: Full per-cohort intra-cohort results on HEST-1k (Pearson mean_{std} over three seeds; nested-validation). All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels, except Mist (V3) which uses UNI (1024-dim) for comparison. HCC (liver), LYMPH_IDC, and READ are excluded per Appendix H. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	Mist (V3)	Mist+CONCH
IDC Breast	0.551±0.01	0.309±0.07	0.567±0.01	0.519±0.00	0.571±0.01	0.551±0.01	0.439±0.01	0.567±0.00	0.581±0.01
COAD Colorectal	0.242±0.01	0.295±0.01	0.285±0.01	0.314±0.01	0.328±0.01	0.293±0.01	0.177±0.01	0.299±0.00	0.318±0.01
CCRCC Kidney	0.163±0.02	0.184±0.02	0.256±0.02	0.175±0.02	0.210±0.02	0.223±0.02	0.123±0.00	0.265±0.02	0.252±0.01
LUNG Lung	0.518±0.01	0.470±0.01	0.587±0.01	0.562±0.01	0.598±0.00	0.584±0.01	0.423±0.03	0.589±0.01	0.589±0.01
PAAD Pancreas	0.359±0.03	0.247±0.02	0.404±0.04	0.419±0.03	0.413±0.04	0.415±0.03	0.233±0.02	0.392±0.04	0.437±0.03
PRAD Prostate	0.294±0.02	0.333±0.02	0.384±0.02	0.348±0.01	0.413±0.01	0.377±0.03	0.207±0.02	0.390±0.02	0.410±0.01
SKCM Skin	0.665±0.01	0.522±0.03	0.653±0.01	0.645±0.01	0.654±0.01	0.631±0.01	0.491±0.04	0.696±0.00	0.707±0.01
Average	0.399	0.337	0.448	0.426	0.455	0.439	0.299	0.457	0.470

Mist (V3) uses UNI (1024-dim); all other methods use UNI + CONCH (1536-dim). STEM: corrected DiT (T=1000, 34M params).

Table 17: Cross-organ LOOO transfer, expanded baselines, per-cohort Pearson $\uparrow$ (HEST-1k) (Pearson mean_{std} over three seeds; nested-validation; liver excluded). Leave-one-organ-out: trained on all other organ groups, tested on the held-out cohort. All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	Mist+CONCH
IDC Breast	0.198±0.05	0.458±0.01	0.465±0.02	0.529±0.02	0.437±0.04	0.567±0.02	0.420±0.03	0.520±0.02
COAD Colorectal	0.420±0.06	0.670±0.03	0.669±0.03	0.662±0.01	0.657±0.05	0.728±0.01	0.452±0.02	0.729±0.03
CCRCC Kidney	0.068±0.01	0.055±0.01	0.106±0.01	0.095±0.01	0.081±0.03	0.103±0.02	0.090±0.01	0.109±0.01
LUNG Lung	0.493±0.07	0.378±0.18	0.657±0.01	0.624±0.01	0.665±0.01	0.672±0.01	0.512±0.03	0.675±0.01
PAAD Pancreas	0.459±0.02	0.399±0.02	0.463±0.00	0.537±0.02	0.513±0.03	0.572±0.01	0.409±0.03	0.548±0.01
PRAD Prostate	0.079±0.00	0.098±0.02	0.114±0.03	0.118±0.01	0.087±0.02	0.107±0.01	0.046±0.00	0.118±0.01
SKCM Skin	0.520±0.16	0.736±0.01	0.725±0.03	0.748±0.01	0.740±0.01	0.750±0.02	0.631±0.03	0.728±0.04
Average	0.319	0.399	0.457	0.473	0.454	0.500	0.366	0.490

F Published Per-Cohort Comparison

G Gene Panel Comparison

Table 20 reports intra-cohort mean PCC averaged over seven cohorts and three seeds for three gene-panel strategies: highly variable genes (HVG, the HEST-1k default), unsupervised differentially expressed genes (DEG; Leiden clustering + Wilcoxon), and histology-morphology HVGs (HMHVG; genes with highest max Pearson correlation to UNI+CONCH PCA components). All methods use identical UNI+CONCH features, nested-validation, and the same training protocol. DEG and HMHVG panels consistently outperform HVG across all methods, with HMHVG yielding the highest mean for every method tested. Mist leads on all three panels, with its advantage most pronounced on HMHVG (0.657 vs. 0.649 for HyperST) and consistent across DEG (0.624 vs. 0.621) and HVG (0.471 vs. 0.449).

Table 21 gives the per-cohort breakdown. Mist leads on 5 of 7 cohorts under HVG, 5 of 7 under DEG, and 5 of 7 under HMHVG. HyperST is stronger on CCRCC (HVG), COAD/IDC (DEG), and IDC/LUNG (HMHVG).

H HEST-1k Cohort Statistics

Table 22 lists the number of samples, total spots, and cross-validation folds for each HEST-1k cohort. Two organ groups in HEST-1k contain multiple cohorts: breast (IDC + LYMPH_IDC) and colorectal (COAD + READ). Within each group the constituent cohorts differ substantially in size: IDC has $2.25\times$ more spots than LYMPH_IDC (44,974 vs. 19,964), and COAD has $2.2\times$ more spots than READ (18,523 vs. 8,407). To avoid misrepresentation from unweighted averaging over unequal cohorts, Table 2 uses IDC as the representative breast cohort and COAD as the representative colorectal cohort, discarding the smaller partner cohorts (LYMPH_IDC and READ) from the organ-level rows. This choice maximises statistical reliability of each organ-level estimate.

Table 18: Cross-organ LOOO transfer with UNI features (Pearson mean_{std} over three seeds; nested-validation; HCC, LYMPH IDC, and READ excluded). Leave-one-organ-out: trained on all other organs, tested on the held-out organ. Best per row in bold.

Organ	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	Mist (V3)
IDC Breast	0.217±0.06	0.459±0.02	0.501±0.02	0.509±0.03	0.452±0.01	0.531±0.03	0.406±0.01	0.534±0.02
COAD Colorectal	0.430±0.03	0.688±0.04	0.669±0.06	0.667±0.02	0.672±0.03	0.749±0.01	0.504±0.03	0.703±0.01
CCRCC Kidney	0.084±0.03	0.084±0.01	0.108±0.01	0.100±0.02	0.122±0.01	0.106±0.01	0.090±0.01	0.124±0.00
LUNG Lung	0.504±0.04	0.499±0.06	0.650±0.01	0.621±0.02	0.667±0.01	0.666±0.01	0.561±0.01	0.660±0.01
PAAD Pancreas	0.404±0.05	0.360±0.07	0.397±0.03	0.521±0.02	0.522±0.01	0.551±0.02	0.460±0.02	0.507±0.04
PRAD Prostate	0.071±0.01	0.104±0.02	0.092±0.03	0.111±0.00	0.107±0.02	0.098±0.02	0.068±0.01	0.119±0.02
SKCM Skin	0.402±0.08	0.652±0.08	0.720±0.02	0.745±0.01	0.724±0.02	0.749±0.01	0.686 [†]	0.692±0.06
Average	0.302	0.407	0.448	0.468	0.466	0.493	0.396 [†]	0.477

[†]STEM DiT (corrected, T=1000, 100 epochs) LOOO: SKCM (skin) has only 1 seed complete; all other organs use 3 seeds. Average marked [†] accordingly.

Table 19: Published per-cohort accuracy on the standard HEST-1k splits, mean PCC[†] as reported in respective papers (ten cohorts, no organ grouping). Shown for context only—not comparable to Table 2, which uses organ grouping and nested-validation. [†]Mist uses our organ-grouped splits, not the standard per-cohort split.

Method	HEST-1k (10 cohorts)
ST-Net He et al. (2020) (NBE'20)	0.286
Hist2ST Zeng et al. (2022) (BIB'22)	—
HyperST Zhang et al. (2026) (CVPR'26)	—
MCToGene Sun et al. (2026) (CVPR'26)	0.435
Mist [†] (ours)	0.440

Table 20: Gene panel comparison: intra-cohort mean PCC[†] (HEST-1k). Mean_{std} over three seeds; per-seed value is the average over seven cohorts. All methods use identical frozen UNI + CONCH features (1536-dim) and nested-validation. Best per column in bold.

Method	HVG	DEG	HMHVG
ST-Net He et al. (2020)	0.398 _{0.004}	0.523 _{0.004}	0.550 _{0.008}
Hist2ST Zeng et al. (2022)	0.330 _{0.032}	0.494 _{0.014}	0.544 _{0.038}
HyperST Zhang et al. (2026)	0.449 _{0.003}	0.621 _{0.002}	0.649 _{0.002}
BLEEP Xie et al. (2023)	0.427 _{0.004}	0.571 _{0.005}	0.595 _{0.014}
STEM Zhu et al. (2025)	0.291 _{0.005}	0.456 _{0.002}	0.485 _{0.010}
Mist (ours)	0.471_{0.005}	0.624_{0.002}	0.657_{0.006}

Table 21: Per-cohort gene panel comparison: intra-cohort PCC $\uparrow$ (HEST-1k). Mean_{std} over three seeds. All methods use UNI + CONCH (1536-dim) and nested-validation. Best per cohort per panel in bold.

Cohort	ST-Net	Hist2ST	HyperST	BLEEP	STEM	Mist
Panel: HVG (highly variable genes, HEST-1k default)						
IDC Breast	0.551 _{0.007}	0.305 _{0.108}	0.568 _{0.006}	0.519 _{0.002}	0.434 _{0.002}	0.581_{0.008}
COAD Colorectal	0.235 _{0.005}	0.298 _{0.009}	0.294 _{0.015}	0.322_{0.006}	0.177 _{0.014}	0.318 _{0.008}
CCRCC Kidney	0.163 _{0.015}	0.180 _{0.030}	0.256_{0.020}	0.175 _{0.015}	0.118 _{0.006}	0.252 _{0.007}
LUNG Lung	0.521 _{0.010}	0.430 _{0.050}	0.585 _{0.005}	0.565 _{0.008}	0.407 _{0.003}	0.589_{0.006}
PAAD Pancreas	0.366 _{0.032}	0.248 _{0.038}	0.398 _{0.033}	0.412 _{0.034}	0.233 _{0.019}	0.437_{0.028}
PRAD Prostate	0.287 _{0.023}	0.323 _{0.012}	0.380 _{0.021}	0.346 _{0.007}	0.207 _{0.016}	0.410_{0.006}
SKCM Skin	0.661 _{0.006}	0.523 _{0.003}	0.661 _{0.010}	0.650 _{0.004}	0.461 _{0.018}	0.707_{0.007}
Mean	0.398	0.330	0.449	0.427	0.291	0.471
Panel: DEG (differentially expressed genes; unsupervised Leiden + Wilcoxon)						
IDC Breast	0.697 _{0.010}	0.549 _{0.067}	0.776_{0.011}	0.731 _{0.014}	0.684 _{0.020}	0.774 _{0.009}
COAD Colorectal	0.497 _{0.003}	0.608 _{0.010}	0.679_{0.004}	0.639 _{0.016}	0.534 _{0.014}	0.677 _{0.003}
CCRCC Kidney	0.312 _{0.014}	0.335 _{0.013}	0.398 _{0.012}	0.304 _{0.003}	0.253 _{0.009}	0.402_{0.013}
LUNG Lung	0.566 _{0.006}	0.518 _{0.006}	0.660 _{0.001}	0.621 _{0.006}	0.440 _{0.009}	0.662_{0.002}
PAAD Pancreas	0.425 _{0.032}	0.352 _{0.011}	0.509 _{0.023}	0.487 _{0.029}	0.306 _{0.042}	0.518_{0.026}
PRAD Prostate	0.413 _{0.004}	0.437 _{0.008}	0.535 _{0.018}	0.466 _{0.005}	0.355 _{0.026}	0.539_{0.012}
SKCM Skin	0.752 _{0.009}	0.656 _{0.004}	0.788 _{0.005}	0.746 _{0.026}	0.617 _{0.023}	0.794_{0.002}
Mean	0.523	0.494	0.621	0.571	0.456	0.624
Panel: HMHVG (histology-morphology HVGs; max Pearson with UNI+CONCH PCA)						
IDC Breast	0.724 _{0.014}	0.633 _{0.082}	0.775_{0.011}	0.739 _{0.018}	0.687 _{0.008}	0.774 _{0.011}
COAD Colorectal	0.503 _{0.010}	0.659 _{0.018}	0.691 _{0.003}	0.645 _{0.004}	0.528 _{0.032}	0.698_{0.005}
CCRCC Kidney	0.339 _{0.007}	0.387 _{0.010}	0.432 _{0.006}	0.349 _{0.010}	0.290 _{0.013}	0.454_{0.017}
LUNG Lung	0.568 _{0.005}	0.598 _{0.003}	0.698_{0.006}	0.661 _{0.002}	0.517 _{0.011}	0.693 _{0.004}
PAAD Pancreas	0.441 _{0.032}	0.341 _{0.038}	0.500 _{0.031}	0.493 _{0.023}	0.305 _{0.027}	0.524_{0.023}
PRAD Prostate	0.452 _{0.002}	0.477 _{0.011}	0.552 _{0.029}	0.432 _{0.045}	0.337 _{0.016}	0.555_{0.022}
SKCM Skin	0.824 _{0.029}	0.710 _{0.141}	0.897 _{0.001}	0.849 _{0.018}	0.731 _{0.060}	0.901_{0.002}
Mean	0.550	0.544	0.649	0.595	0.485	0.657

Table 22: HEST-1k cohort statistics. Spots counted after barcode filtering. Folds = number of train/test splits in the per-cohort cross-validation. HCC (liver) is excluded from Table 2 due to its near-zero signal floor.

Cohort	Organ	Samples	Spots	Folds
IDC	Breast	4	44,974	4
LYMPH_IDC	Breast	4	19,964	4
COAD	Colorectal	4	18,523	2
READ	Colorectal	4	8,407	2
CCRCC	Kidney	24	74,220	6
HCC	Liver*	2	4,248	2
LUNG	Lung	2	7,505	2
PAAD	Pancreas	3	9,793	3
PRAD	Prostate	23	62,710	2
SKCM	Skin	2	5,716	2

*HCC (liver) is included here for completeness but excluded from Table 2; all baselines score near zero on this cohort due to low spatial gene-expression variance.

I Proofs of the formal guarantees

I.1 Proof of Proposition 1

For every pair of spots i, j , orthogonality of R gives

$$\|R\mathbf{p}_i + \mathbf{t} - (R\mathbf{p}_j + \mathbf{t})\|_2 = \|R(\mathbf{p}_i - \mathbf{p}_j)\|_2 \quad (9)$$

$$= \|\mathbf{p}_i - \mathbf{p}_j\|_2. \quad (10)$$

Hence, transforming the coordinate frame leaves the complete distance matrix unchanged. It also leaves the k NN graph, radial basis features, and distance-dependent attention biases unchanged. The input projection is unchanged because $\mathbf{F}$ is fixed. At every layer, the local pathway therefore receives the same features, neighbours, and distance biases, while the global and slide pathways do not use coordinates. Their outputs, the residual fusion, and the feed-forward update are identical. Induction over the L layers gives identical final representations and predictions, proving Proposition 1.

I.2 Proof of Proposition 2

Let $\mathbf{F}' = Q\mathbf{F}$ and $\mathbf{P}' = Q\mathbf{P}$. The pairwise distance matrix is relabelled as $\mathbf{D}' = Q\mathbf{D}Q^\top$. Under equivariant tie handling, the k NN graph and its radial basis edge features are relabelled by the same permutation. The input projection and all pointwise linear maps, layer normalizations, activations, and feed-forward networks commute with Q .

For local attention, permuting the spots only relabels each neighbourhood and its attention sum, so the local output is multiplied by Q . Global self-attention is likewise permutation equivariant because its query-key score matrix is transformed by Q on both axes and its value matrix by Q . The slide attention scores are permuted, but their weighted sum is unchanged; broadcasting that same summary to all spots is equivariant. Consequently, residual fusion preserves equivariance at every layer. Induction over the layers and application of the pointwise prediction head yield $f_\theta(Q\mathbf{F}, Q\mathbf{P}) = Qf_\theta(\mathbf{F}, \mathbf{P})$, proving Proposition 2.

J Three-Metric Intra-Cohort Summary

Table 23 extends the intra-cohort comparison with Spearman correlation (SCC) and mean squared error (MSE) alongside PCC, averaged over seven organ groups.

Table 23: Intra-cohort three-metric summary (HEST-1k, averaged over seven organ groups; three seeds; UNI + CONCH features). PCC $\uparrow$, SCC $\uparrow$, MSE $\downarrow$. Best per column in bold.

Method	PCC $\uparrow$	SCC $\uparrow$	MSE $\downarrow$
ST-Net He et al. (2020) (NBE'20)	0.399	0.356	2.043
Hist2ST Zeng et al. (2022) (BIB'22)	0.337	0.321	1.616
HyperST Zhang et al. (2026) (CVPR'26)	0.448	0.411	1.439
BLEEP Xie et al. (2023) (NeurIPS'23)	0.426	0.388	1.464
STEM Zhu et al. (2025) (ICLR'25)	0.299	0.274	1.855
Mist (ours)	0.470	0.430	1.389

K Per-Cohort MSE and MAE (Intra-Cohort)

Tables 24 and 25 report the per-cohort mean squared error (MSE) and mean absolute error (MAE) for the same models, cohorts, and protocol as the intra-cohort Pearson comparison in Table 2. Mist attains the best average on both error metrics (MSE 1.389, MAE 0.829), consistent with its Pearson result; per-cohort leadership is less uniform than on Pearson, as the training objective couples MSE with a correlation term.

Table 24: Intra-cohort MSE $\downarrow$ on HEST-1k, per cohort. Mean_{std} over three seeds with nested-validation; best (lowest) per row in bold. Same models, cohorts, features (frozen UNI + CONCH, 1536-dim), 50-gene panels, and training budget as Table 2.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	STEM ICLR'25	Mist+CONCH
IDC Breast	2.966 $\pm$ 0.311	2.740 $\pm$ 0.194	2.331 $\pm$ 0.065	2.503 $\pm$ 0.061	2.811 $\pm$ 0.056	2.242$\pm$0.031
COAD Colorectal	2.335 $\pm$ 0.063	2.195$\pm$0.058	2.244 $\pm$ 0.046	2.201 $\pm$ 0.062	2.632 $\pm$ 0.021	2.322 $\pm$ 0.031
CCRCC Kidney	0.587 $\pm$ 0.007	0.578 $\pm$ 0.032	0.527$\pm$0.016	0.604 $\pm$ 0.011	0.763 $\pm$ 0.016	0.556 $\pm$ 0.019
LUNG Lung	3.388 $\pm$ 0.062	1.732 $\pm$ 0.003	1.616$\pm$0.009	1.650 $\pm$ 0.024	2.036 $\pm$ 0.052	1.665 $\pm$ 0.043
PAAD Pancreas	2.051 $\pm$ 0.235	1.331 $\pm$ 0.193	1.215 $\pm$ 0.237	1.151$\pm$0.258	1.711 $\pm$ 0.206	1.173 $\pm$ 0.166
PRAD Prostate	0.586 $\pm$ 0.011	0.560 $\pm$ 0.020	0.568 $\pm$ 0.017	0.570 $\pm$ 0.019	0.822 $\pm$ 0.014	0.537$\pm$0.034
SKCM Skin	2.387 $\pm$ 0.085	2.176 $\pm$ 0.016	1.573 $\pm$ 0.078	1.570 $\pm$ 0.036	2.207 $\pm$ 0.260	1.227$\pm$0.030
Average	2.043	1.616	1.439	1.464	1.855	1.389

Table 25: Intra-cohort MAE $\downarrow$ on HEST-1k, per cohort. Mean_{std} over three seeds with nested-validation; best (lowest) per row in bold. Same models, cohorts, features (frozen UNI + CONCH, 1536-dim), 50-gene panels, and training budget as Table 2.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	STEM ICLR'25	Mist+CONCH
IDC Breast	1.273 $\pm$ 0.066	1.246 $\pm$ 0.032	1.120 $\pm$ 0.016	1.141 $\pm$ 0.022	1.210 $\pm$ 0.015	1.096$\pm$0.010
COAD Colorectal	0.996 $\pm$ 0.014	1.008 $\pm$ 0.017	1.021 $\pm$ 0.008	0.986$\pm$0.020	1.088 $\pm$ 0.008	1.040 $\pm$ 0.012
CCRCC Kidney	0.575 $\pm$ 0.003	0.574 $\pm$ 0.018	0.559$\pm$0.010	0.583 $\pm$ 0.003	0.612 $\pm$ 0.011	0.569 $\pm$ 0.008
LUNG Lung	1.373 $\pm$ 0.014	0.985 $\pm$ 0.002	0.944$\pm$0.003	0.960 $\pm$ 0.008	1.050 $\pm$ 0.003	0.957 $\pm$ 0.014
PAAD Pancreas	1.013 $\pm$ 0.051	0.872 $\pm$ 0.055	0.806 $\pm$ 0.075	0.796$\pm$0.088	0.936 $\pm$ 0.054	0.798 $\pm$ 0.046
PRAD Prostate	0.564 $\pm$ 0.004	0.546$\pm$0.003	0.559 $\pm$ 0.028	0.547 $\pm$ 0.007	0.620 $\pm$ 0.014	0.550 $\pm$ 0.017
SKCM Skin	1.015 $\pm$ 0.023	1.056 $\pm$ 0.015	0.939 $\pm$ 0.027	0.926 $\pm$ 0.011	1.049 $\pm$ 0.070	0.791$\pm$0.005
Average	0.973	0.898	0.850	0.848	0.938	0.829

L Cross-Organ Generalization: PCC and MSE

Table 26 reports both PCC and MSE for the LOOO and POOLED cross-organ regimes, complementing the PCC-only summaries in the main paper.

Table 26: Cross-organ generalization: PCC and MSE (three seeds, mean_{std}). looo: leave-one-organ-out transfer; pooled: 5-fold pooled cross-validation. PCC $\uparrow$ / MSE $\downarrow$ on HEST-1k.

Method	looo		pooled	
	PCC $\uparrow$	MSE $\downarrow$	PCC $\uparrow$	MSE $\downarrow$
ST-Net He et al. (2020) (NBE'20)	0.267 _{0.02}	1.846 _{0.10}	0.370 _{0.01}	0.907 _{0.05}
Hist2ST Zeng et al. (2022) (BIB'22)	0.360 _{0.02}	1.182 _{0.07}	0.737 _{0.01}	0.488 _{0.01}
HyperST Zhang et al. (2026) (CVPR'26)	0.395 _{0.01}	1.136 _{0.10}	0.756 _{0.01}	0.465 _{0.02}
BLEEP Xie et al. (2023) (NeurIPS'23)	0.416 _{0.01}	1.010_{0.02}	0.758 _{0.01}	0.427_{0.02}
STEM Zhu et al. (2025) (ICLR'25)	0.322 _{0.01}	1.274 _{0.00}	0.643 _{0.01}	0.594 _{0.01}
Mist (ours)	0.421_{0.01}	1.233 _{0.11}	0.760_{0.01}	0.440 _{0.02}

M Intra-Cohort Results with UNI + CONCH Features, Extended Baselines

Table 27 extends the intra-cohort comparison to include M2OST and EGN under the same UNI + CONCH feature protocol.

Table 27: Intra-cohort results using UNI+CONCH features, extended baselines (Pearson mean_{std} over three seeds; nested-validation). All methods use identical frozen UNI+CONCH features (1536-dim) and 50-gene panels. HCC (liver), LYMPH_IDC, and READ are excluded per Appendix H. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	Mist (ours)
IDC Breast	0.551±0.01	0.309±0.06	0.567±0.01	0.519±0.00	0.571±0.01	0.551±0.01	0.439±0.01	0.581±0.01
COAD Colorectal	0.242±0.01	0.295±0.01	0.285±0.01	0.314±0.01	0.328±0.01	0.293±0.01	0.177±0.01	0.318±0.01
CCRCC Kidney	0.163±0.02	0.184±0.02	0.256±0.02	0.175±0.02	0.210±0.02	0.223±0.02	0.123±0.00	0.252±0.01
LUNG Lung	0.518±0.01	0.470±0.01	0.587±0.00	0.562±0.01	0.598±0.00	0.584±0.01	0.423±0.03	0.589±0.01
PAAD Pancreas	0.359±0.03	0.247±0.02	0.404±0.04	0.419±0.03	0.413±0.04	0.415±0.03	0.233±0.02	0.437±0.03
PRAD Prostate	0.294±0.02	0.333±0.02	0.384±0.02	0.348±0.01	0.413±0.01	0.377±0.03	0.207±0.02	0.410±0.01
SKCM Skin	0.665±0.01	0.522±0.03	0.653±0.01	0.645±0.01	0.654±0.01	0.631±0.01	0.491±0.04	0.707±0.01
Average	0.399	0.337	0.448	0.426	0.455	0.439	0.299	0.470

All entries use three completed seeds. STEM uses the corrected DiT configuration ($T = 1000$, 100 epochs).

N Training Objective Ablation

Table 28 isolates the contribution of the Pearson correlation term in the training objective from the architectural design, confirming that the accuracy gap over baselines is not attributable to the loss function alone.

Table 28: Objective ablation (HEST-1k). Effect of adding the Pearson correlation term to the MSE objective; averaged over seven organ groups, three seeds.

Model	Objective	PCC ↑
Mist	MSE only	0.382
Mist	MSE + PCC	0.389